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APEX RETAIL GROUP | ANALYTICS TRAINING PROGRAM
DOCUMENT: CLASS NOTES (SESSION 8)
TOPIC: MID-COURSE CHECKPOINT — PRACTICAL EXPLORATORY DATA ANALYSIS (EDA)
LEVEL: INTERMEDIATE-ADVANCED (DAY 8)

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1. CONTINUITY & BUSINESS SCENARIO RECAP

Over the past seven sessions, you built your core technical and analytical foundation:
- Session 1: Translating business questions, variables, lists, and basic metrics (`sum`, `len`).
- Session 2: Conditional logic (`if`/`else`), loops, functions, and error handling (`try`/`except`).
- Session 3: Vectorized array operations and dispersion analysis (`std`, `var`) with NumPy.
- Session 4: Loading, structural inspection (`.shape`, `.head()`), and metadata auditing (`.info()`).
- Session 5: Cleaning messy data (deduplication, string stripping/title-casing, `.fillna()`, `to_datetime`).
- Session 6: Aggregating and segmenting business drivers using `.groupby()`, `.agg()`, and `.sort_values()`.
- Session 7: Relational wrangling across separate files using `pd.merge()` (Left/Inner joins).

Now, in your eighth week at Apex Retail Group:

Victoria Vance, Vice President of Retail Operations, requests your first major independent investigation:

> **"Apex Retail Group experienced a alarming drop in Q3 store operating profitability. Executive leadership needs to know why.**
>
> **We are providing you with two raw, uncleaned exports:**
> **1. Q3 Transaction Sales Logs (CSV file with missing values, string defects, and duplicate entries).**
> **2. Store Facility Metadata (CSV file containing store region, square footage, and operational cost tiers).**
>
> **You must execute a full, end-to-end Exploratory Data Analysis (EDA) without step-by-step prompts: audit the data, clean and sanitize both files, execute a Left Join to combine them, isolate where revenue/margin leakage occurred, and write a formal executive briefing note with actionable recommendations."**

In this session, you consolidate all tools learned thus far into a unified, end-to-end Exploratory Data Analysis (EDA) workflow.

2. THE EXPLORATORY DATA ANALYSIS (EDA) WORKFLOW

EDA is the iterative process where an analyst inspects, cleans, transforms, and interrogates a dataset to discover patterns, isolate anomalies, test hypotheses, and check assumptions.

THE 5-STAGE EDA PIPELINE
1. DATA AUDIT : Inspect `.shape`, `.info()`, `.describe()`, identify defects
2. SANITIZATION : Deduplicate, fix dtypes, clean strings, impute `NaNs`
3. RELATIONAL MERGE : Join transaction logs with store metadata via shared key
4. INVESTIGATION : Multi-dimensional `.groupby()` across regions & categories
5. SYNTHESIS : Translate findings into business implications & advice

3. CORE EDA SYNTAX CONSOLIDATION

A. Data Auditing & Sanitization

```python

import pandas as pd
import numpy as np

# Audit geometry & nulls
print(df.shape)
print(df.isnull().sum())

# Deduplicate & clean strings
df = df.drop_duplicates().reset_index(drop=True)
df["category"] = df["category"].str.strip().str.title()

# Date casting & median imputation
df["order_date"] = pd.to_datetime(df["order_date"])
median_val = df["revenue"].median()
df["revenue"] = df["revenue"].fillna(median_val)
```

B. Relational Merging & Derived Metric Engineering

```Python

# Left join transaction data with facility metadata
df_merged = pd.merge(df_txns, df_stores, on="store_id", how="left")

# Feature engineering: Calculating Net Profit ($) and Profit Margin (%)
df_merged["net_profit"] = df_merged["revenue"] -
df_merged["operating_cost"]
df_merged["margin_pct"] = (df_merged["net_profit"] / df_merged["revenue"])
* 100
```

C. Multi-Dimensional Investigation

```Python

# Aggregating across Store Region and Category
eda_summary = df_merged.groupby(["store_region", "category"]).agg(
    total_revenue=("revenue", "sum"),
    total_profit=("net_profit", "sum"),
    avg_margin=("margin_pct", "mean"),
    order_count=("order_id", "count")
).sort_values(by="total_profit", ascending=True)
```

EXECUTING THE Q3 PROFITABILITY INVESTIGATION

Below is the clean, annotated Python implementation for Victoria's Q3 audit:

```Python

import pandas as pd
import numpy as np

# Step 1: Raw Q3 Transaction Export (Imperfect Data)
raw_txns = {
    "order_id": ["ORD-801", "ORD-802", "ORD-802", "ORD-803", "ORD-804",
"ORD-805", "ORD-806", "ORD-807"],
    "store_id": ["STR-10", "STR-20", "STR-20", "STR-10", "STR-30", "STR-
20", "STR-30", "STR-10"],
    "order_date": ["2026-07-05", "2026-07-12", "2026-07-12", "2026-08-01",
"2026-08-15", "2026-09-02", "2026-09-10", "2026-09-20"],
    "category": [" Electronics ", "Apparel", "Apparel", " ELECTRONICS ",
"Home", " Apparel ", "Home", "Electronics"],
    "revenue": [1500.00, 220.00, 220.00, np.nan, 310.00, 180.00, 290.00,
1200.00],
    "units": [3, 4, 4, 2, 5, 3, 4, 2]
}

# Step 2: Store Metadata
raw_stores = {
    "store_id": ["STR-10", "STR-20", "STR-30"],
    "store_region": ["North", "South", "West"],
    "operating_cost": [400.00, 150.00, 350.00]  # Cost per order fulfilled
}

df_txns = pd.DataFrame(raw_txns)
df_stores = pd.DataFrame(raw_stores)

# Step 3: Sanitize Transactions
df_txns = df_txns.drop_duplicates().reset_index(drop=True)
df_txns["category"] = df_txns["category"].str.strip().str.title()
df_txns["order_date"] = pd.to_datetime(df_txns["order_date"])
df_txns["revenue"] = df_txns["revenue"].fillna(df_txns["revenue"].median())

# Step 4: Relational Merge
df_full = pd.merge(df_txns, df_stores, on="store_id", how="left")

# Step 5: Engineer Profitability Metrics
df_full["net_profit"] = df_full["revenue"] - df_full["operating_cost"]

# Step 6: Multi-Dimensional EDA Aggregation
region_summary = df_full.groupby("store_region").agg(
    total_revenue=("revenue", "sum"),
    total_operating_cost=("operating_cost", "sum"),
    net_profit=("net_profit", "sum"),
    order_count=("order_id", "count")
).sort_values(by="net_profit", ascending=True)

print("--- REGIONAL PROFITABILITY SUMMARY ---")
print(region_summary)
```

INTERPRETING NUMBERS & FORMULATING BUSINESS RECOMMENDATIONS

A. Observation (The Investigation Findings):

- Total Q3 Audited Revenue: $4,900.00 across 7 unique transactions.
- Regional Profitability Breakdown:
    - North Region (STR-10): Revenue = $3,900.00 | Operating Costs = $1,200.00 | Net Profit = +$2,700.00 (High Margin).
    - South Region (STR-20): Revenue = $400.00 | Operating Costs = $300.00 | Net Profit = +$100.00 (Low Volume, Positive Margin).
    - West Region (STR-30): Revenue = $600.00 | Operating Costs = $700.00 | Net Profit = -$100.00 (Net Operating Loss!).

B. Interpretation & Business Meaning:

- Overall top-line revenue ($4.9k) masks severe regional profit leakage in the West Region.
- West Region store STR-30 is losing money on every order processed because its fixed operating cost per fulfillment ($350.00) exceeds its average transaction revenue ($300.00).

C. Professional Recommendation to Victoria Vance (VP of Operations):

"An end-to-end EDA reveals that Q3 profitability decline is concentrated in the West Region (STR-30), which generated a net loss of -$100.00 on $600.00 in sales due to excessive operating costs ($350/order). Conversely, the North Region generated $2,700.00 in net profit.

Recommendation: Immediately review fulfillment overhead at West Region facility STR-30 to lower per-order operating costs, or implement a $400 minimum order threshold for West Region store deliveries."

CORE MINDSET HABITS REINFORCED TODAY

1. Connect top-line metrics to net profit: High revenue means nothing if operating overhead consumes all margin.
2. Execute end-to-end independently: Standardize raw inputs, join metadata, engineer features, and isolate root causes without needing guided instructions.